

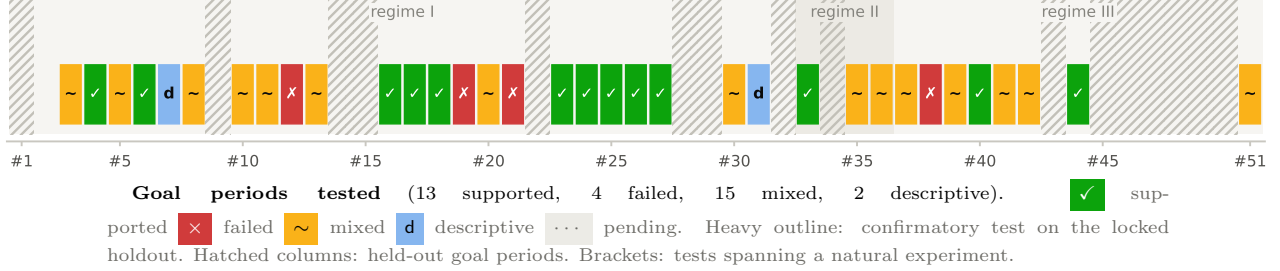
H92: RMT-cleaned matrices forecast tomorrow's alignment

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

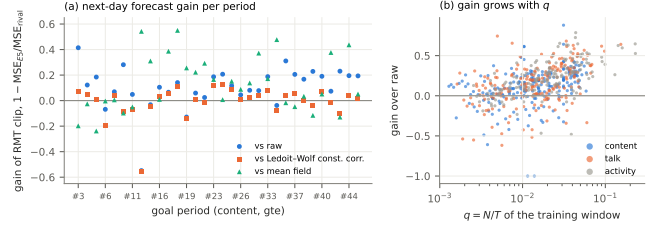
Bottom line. Random-matrix cleaning removes sampling noise from agent correlations as theory predicts, but the persistent structure is one uniform mode, so plain shrinkage does as well.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

The test asks which estimate of today's agent correlation matrix best forecasts tomorrow's. It compares the raw matrix, two Ledoit–Wolf shrinkages, a mean-field matrix and random-matrix clipping, which keeps only eigenmodes above a calibrated noise edge. Forecasts run inside 34 non-holdout goal periods (205 target days in content, both embedding models; talk and activity too). Clipping beats the raw matrix in 26/32 and 27/32 periods (mean gain 0.10 [0.03, 0.16]). The gain grows with $q = N/T$, as random-matrix theory predicts. Clipping does not beat shrinkage: it ties Ledoit–Wolf constant correlation within 2%. Talk pair structure is not forecastable beyond its mean. Cleaning erases the #38 talk room block. The holdout script exists and is not run.



Observables. (a) Gain of clipping over each rival, per goal period (content, gte). It is positive against raw (blue) in 27/32 periods and against mean field (green) in 23/32, and it scatters around zero against constant-correlation shrinkage (orange). (b) Gain over raw for every forecast against $q = N/T$ of the training window. The gain grows with q in content, talk and activity.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To track who will co-move with whom tomorrow, do not use the raw pair correlations. Clip them at the noise edge or shrink them toward the mean correlation: either removes about a tenth of the error, and more when the window is short. Constant-correlation shrinkage is the simpler choice and does as well. Expect little: the best forecast cuts the error of “no correlation” by only 30% in content. In talk the gain is 5% or less. Do not clip talk if room structure matters: in #38 clipping deleted the room block.

PHYSICAL MODEL

Inverse Ising (model 01) with random-matrix cleaning. A spin is an agent's day-centered 30-min content vector, or its talk or activity state per minute in the all-present window. S is the training correlation matrix over all earlier days of the unit, with N agents and T samples. λ_+ is the 95th percentile of the top eigenvalue when agents' series are shifted against each other. Clipping replaces every eigenvalue below λ_+ by their mean. The target is the next day's realized matrix Q .

$$\hat{C}_{\text{clip}} = \sum_{\lambda_k > \lambda_+} \lambda_k v_k v_k^\top + \bar{\lambda}_{\text{bulk}} \sum_{\lambda_k \leq \lambda_+} v_k v_k^\top, \quad \hat{C}_{\text{LW}} = (1 - \alpha)S + \alpha F$$

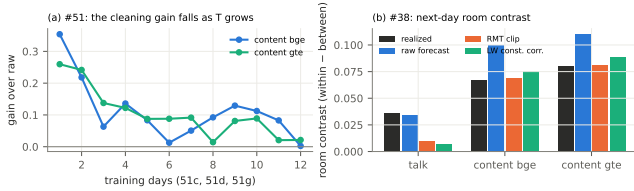
$$\text{MSE} = \langle (\hat{C}_{ij} - Q_{ij}(t+1))^2 \rangle_{i < j}, \quad r = 1 - \frac{\text{MSE}_{\text{clip}}}{\text{MSE}_{\text{rival}}}$$

$$\text{Var}(S_{ij}) \approx \frac{(1 - \rho_{ij}^2)^2}{T}, \quad \lambda_+^{\text{MP}} = (1 + \sqrt{N/T})^2$$

H92-rmt-cleaned-forecast · generated 2026-10-04 13:50 by infra/summaries/build_summaries.py · card hypotheses/H92-rmt-cleaned-forecast/README.md · ratings: H92 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P0 synthetic edge and ordering	spurious modes 3–17%; two-room advantage only at $N \geq 16, W \geq 8$	partial
P1 clip beats raw (content)	26/32, 27/32 periods; 0.10 [0.03, 0.16], 0.11 [0.04, 0.16]	pass
P2 clip beats best shrinkage	17/32 both models; vs const. corr. $-0.020, -0.008$	fail (tie)
P3 channels	talk gain > 0 in 27/30, beats LW-CC 10/30; activity skill 0, raw -0.39	mixed
P4 gain grows with q	Spearman $0.45\text{--}0.65$, every channel	pass
P5 calibrated edge beats MP edge	$k_{\text{MP}} > k_{\text{cal}}$ in 20%; gain $-0.015, 0.000$	fail
P6 longer training	helps raw (79–92%); hurts clip (–8%)	pass
N1 #51 gain falls with T	Spearman $-0.69, -0.90$; beats LW-CC 4/7, 5/7 units	mixed
N2 #38 rooms kept	talk: kept on 2/10 days; content: 0.069 vs real 0.067	fail



Second figure. (a) #51: the gain of clipping over raw falls as the training window grows from 1 to 12 days, as the $1/T$ noise law predicts. (b) #38: next-day room contrast (within minus between rooms). In talk, clipping deletes it. In content, clipping keeps it and is closer to the realized value than raw.

GOAL PERIODS IN DETAIL

G03 ~ mixed
G04 ✓ supported
G05 ~ mixed
G06 ✓ supported
G07 d descriptive
G08 ~ mixed
G10 ~ mixed
G11 ~ mixed
G12 × failed
G13 ~ mixed

+24 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	2	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The target is itself a noisy one-day matrix, so all estimators share a large error floor; only differences are meaningful. The calibrated edge keeps a spurious mode in up to 17% of noise fits. Most periods have $N < 16$, where a real second mode is cut, so P2 is powered only in #42 and #51. Training matrices zero-fill absent agent-days. No natural experiment was tested.

ROUND 2

Add the rotationally invariant estimator and a one-mode-plus-rooms factor model. Use an exponentially weighted training window, since cleaned structure drifts. Report the cleaned top mode per period on the phase diagram.

- **Essence.** how many persistent collective modes the swarm has once sampling noise is removed; the tie with constant-correlation shrinkage says one.
- **H92-R1** Add the rotationally invariant estimator (Ledoit–Péché / Bun–Bouchaud–Potters) and a one-mode-plus-rooms factor model.
- **H92-R2** Forecast with an exponentially weighted training window: cleaned structure drifts, so recency should beat the expanding window.
- **H92-R3** Use the cleaned top mode as the per-period coupling coordinate on the phase diagram.